COA207 – Foundations of Artificial Intelligence
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Domain Chosen: Route Finding

Component 1: Problem Analysis & System Design

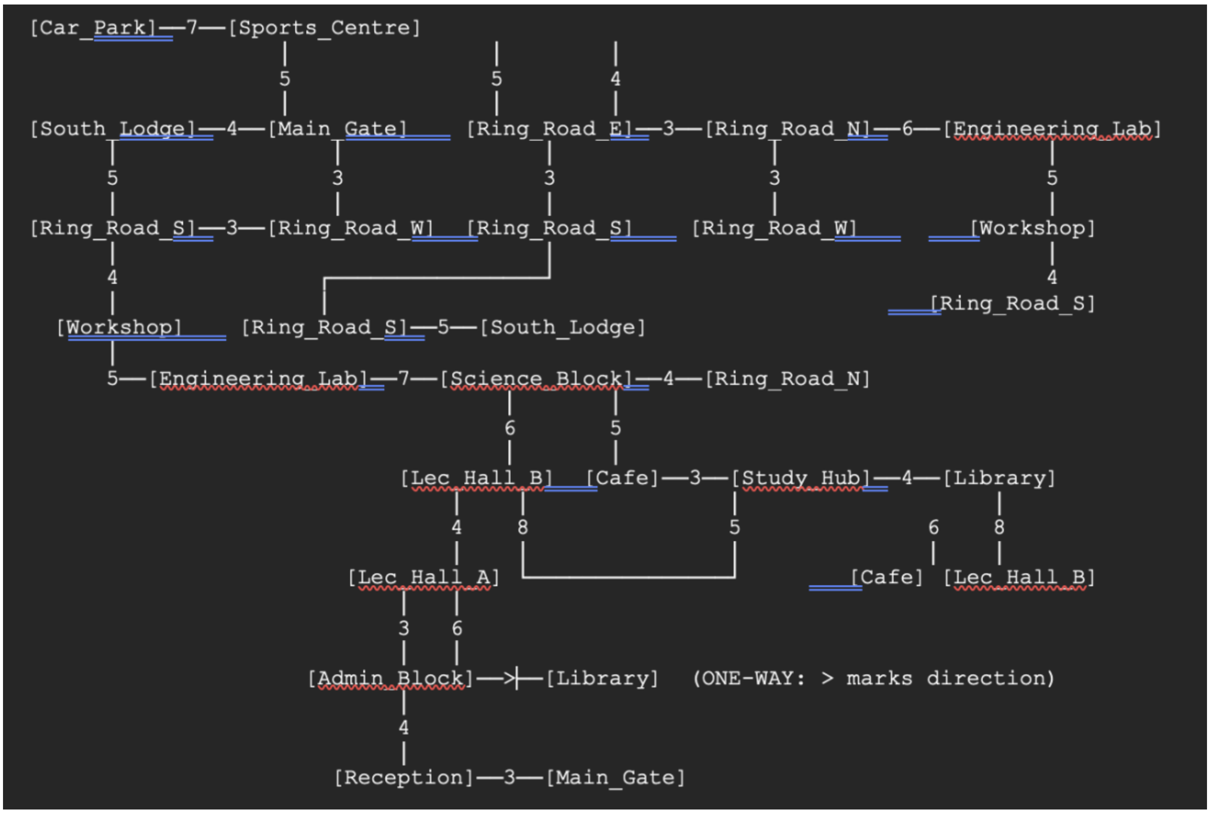
The Edinburgh Campus Emergency Route Finder supports emergency services, facility managers, and first-aiders navigating a fictional university campus. The campus is modelled as a weighted directed graph. 18 locations connected by routes whose costs represent travel time in minutes. Given a start and goal location, the system returns the minimum-cost path subject to real-world constraints that a standard graph search formulation does not capture.

Russell and Norvig (2022, ch. 2) provide the framework for classifying the environment. It is fully observable, and the agent has complete map knowledge before searches begin. It is deterministic and static. Edge costs are fixed integers, and the map does not change during a query. Each query is independent (episodic), and all locations and transitions are discrete.

Four constraints shape the graph. The Admin Block to Library route leads to a one-way fire-exit corridor. Going in the reverse direction requires a detour through the Lecture Halls. The Library Bridge is absent from both endpoints' adjacency lists as it is closed for construction. The Maintenance Road does not appear in the graph as it is restricted to authorised personnel; it is structurally inaccessible rather than merely expensive. The Ring Road runs as a directed cycle (N > E > S > W > N), and the algorithm must exit it without looping. The campus layout is illustrated below.

Network Diagram

Key:
Nodes represent campus locations.
Edge numbers are distances/weights.
> denotes a one-way connection.



Justification of Approach

A* search is the primary algorithm. It is both optimal and complete when paired with an admissible heuristic, defined as one that never overestimates the true remaining cost (Russell & Norvig, 2022, Ch. 3). The heuristic used here consists of pre-computed straight-line time estimates to the goal, set conservatively to preserve this guarantee. Three factors motivate the choice. Edge costs are non-uniform, which disqualifies BFS as a candidate. Spatial heuristics are available, which means settling for UCS would discard useful information. And in an emergency context, expanding fewer nodes has practical value.

Uniform Cost Search is also complete and optimal but operates without a heuristic, expanding nodes in cost order until the goal is reached. Where an admissible heuristic exists, A* will expand no more nodes and typically far fewer. On the primary test case here, A* expanded 6 nodes against UCS's 17. UCS is implemented for that comparison.

BFS fits the criteria of the weakest of the three for this scenario as it minimises hop count and not cost. This means it could return a two-hop, thirty-minute path in preference to a three-hop, twelve-minute one. For emergency routing, that trade-off is unacceptable, and BFS is included only to make the contrast concrete.

System Design

The campus is represented as a Python dictionary of dictionaries. Constraints are encoded in the graph structure rather than handled at runtime. One-way routes appear only in the source node's entry. Blocked routes are absent from both endpoints. The restricted road has no entry at all. Any path the algorithm cannot see, it cannot take. A* maintains a min-heap frontier ordered by f(n) = g(n) + h(n). An explored dictionary tracks visited nodes and supports lazy deletion of stale frontier entries.

```
frontier ← min-heap [ (h(s), 0, s, [s]) ]   // (f, g_cost, state, path)
explored ← {}                               // state → best g_cost seen

LOOP:
  if frontier EMPTY → return FAILURE
  (f, g, state, path) ← POP-MIN(frontier)
  if explored[state] ≤ g → SKIP            // stale entry
  explored[state] ← g
  if state == goal → return (path, g)
  FOR each (neighbour, cost) in G[state]:
    new_g ← g + cost
    new_f ← new_g + h(neighbour)
    if neighbour not in explored OR explored[neighbour] > new_g:
      PUSH(frontier, (new_f, new_g, neighbour, path + [neighbour]))
```

Component 2: Implementation

Component 3: Reflection & Ethical Analysis

The implementation follows the design set out in Component 1. The most significant decision was encoding all constraints as a graph structure rather than procedural checks. This kept each algorithm general-purpose and placed campus access rules at the data level, where they are easy to inspect and modify. On the primary test case, A* expanded 6 nodes against UCS's 17, a 65% reduction that confirms the heuristic is doing useful work. Agreement between the two algorithms across all seven tests confirms that the admissibility condition holds, satisfying the optimality guarantee from Russell and Norvig (2022, ch. 3).

Three limitations stand out. Firstly, edge costs are fixed integers. Real travel times shift with pedestrian density, weather, and gate access delays, none of which the current model captures. Second, the heuristic is precomputed for a single goal node. Queries to other destinations use zero-heuristic A*, which is still correct but identical in behaviour to UCS. Lastly, no test case exercises a disconnected query. The algorithm returns (None) correctly, but that is not a usable output for emergency personnel, and a real deployment would need to handle it explicitly.

Contrast with a Neural Route Planner

Google Maps-style learned planners (Derrow-Pinion et al., 2021) work differently: travel times are predicted from historical GPS and sensor data rather than read off a static graph. This means costs can update as conditions change, the system generalises across destinations without per-goal heuristics, and it scales to city-level graphs. For a small campus with fixed, enumerable rules, none of that is necessary. The classical approach is easier to verify, easier to explain, and its behaviour under any input can be inspected directly. For emergency use, that auditability matters.

Ethical Analysis

Transparency is where the ethical gap is sharpest. The A* system produces a fully auditable decision trace: every node expanded, every cost accumulated, and the exact reason a path was chosen over alternatives. A neural planner offers none of this by default. Post-hoc methods such as SHAP values or attention maps gesture at its internal reasoning rather than represent it (Rudin, 2019). In most applications, that is a tolerable limitation. In emergency routing, where a first aider needs to trust the output enough to act on it immediately, it is not. A system that cannot be interrogated transfers its uncertainty onto the people following its instructions.

Accountability compounds this problem in a specific way. When the rule-based system fails, the cause is locatable: a wrong entry in the dictionary, correctable at source. Neural errors may stem from spurious correlations in training data that never surface without exhaustive testing, and responsibility for them is distributed across data collectors, model developers, deploying institutions, and operators with no clear point of correction (Mittelstadt et al., 2016). Better models do not resolve that. It is a structural property of how such systems are built and deployed.

Bias also behaves differently across the two systems. The rule-based system applies the same graph to every user. Any structural inequality, say a poorly connected area of campus, sits visibly in the adjacency dictionary and can be fixed by editing it. A neural system trained on historical data risks replicating patterns from that history, including ones that were discriminatory. If emergency services underserved certain campus areas in the past, the model may learn to do the same, in ways that are genuinely difficult to detect (Barocas et al., 2023).

Data dependency raises a further concern. The system here requires no personal data: the graph is static, manually constructed, and public. A neural alternative would need GPS traces, crowd density feeds, and access control logs, all carrying GDPR obligations and realistic risks of mission creep (European Parliament, 2016). Data infrastructure, once built for one purpose, tends not to stay confined to it.

Predictability matters most under stress. The rule-based system is fully deterministic: same input, same output, every time. A neural planner trained on normal conditions can fail silently on inputs outside that distribution, returning a confident recommendation with no indication that it is wrong. Parasuraman and Riley (1997) identified over-reliance as a central failure mode of human-automation systems; uncalibrated confidence makes it worse.

The neural approach is not without genuine advantages. Adaptability, scalability, and real-time responsiveness are real, and the rule-based system has none of them. But Rudin (2019) argues that interpretable models should be the default for high-stakes decisions unless a black-box system offers an accuracy gain that cannot otherwise be achieved. A well-defined 18-node campus graph does not come close to that threshold.

Conclusion

A* search, implemented alongside UCS and BFS for comparison, correctly handles four non-trivial graph constraints, returns optimal paths across all seven test cases, and does so with substantially less node expansion than uninformed alternatives. The comparison with neural route planners was not included to dismiss them. It was included because the choice of method in a safety-critical context is not purely technical. Transparency, accountability, and predictable failure matter. For this system, A* provides all three. The neural alternative does not, and no capability advantage it offers compensates for that on a campus emergency network.

References

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